

Yacin Ben Kacem

Software Engineering Student — Agentic AI & Backend Systems

yacinbenkacem19@gmail.com | +216 29 504 806 | yacinbenkacem.me | linkedin.com/in/yacin-ben-kacem | github.com/YACINBK

SUMMARY

Second-year Software Engineering student at ENISO focused on agentic AI, secure backend orchestration, and automation pipelines built for real integration constraints. Connects LLMs, RAG, OCR, and tool-calling agents to production-minded backends with identity, caching, containerization, and fallback-aware delivery.

EXPERIENCE

White Cape Technologies — AI & Automation Intern

Sousse, Tunisia

Jun – Aug 2026

Agentic RAG Knowledge Assistant

- Architected a **9-node LangGraph state machine** — classify → retrieve → rerank → relevance gate → generate → faithfulness check — returning grounded, cited, role-aware answers streamed over SSE.
- Built a **fully local inference stack** (LiteLLM → Ollama, Qdrant, BGE-M3 embeddings, bge-reranker-v2-m3 via Hugging Face TEI) so no document content ever leaves the network.
- Enforced a **two-dimension privilege filter** (role + admin scope) inside the retrieval query rather than the UI, behind Keycloak OIDC session auth with CSRF protection.
- Gated every response on a faithfulness check that falls back honestly instead of hallucinating; delivered with Celery ingestion, Qdrant semantic caching, and **218 contract-driven tests**.

UX Insight Platform

Jun – Aug 2025

- Delivered an AI-driven **distributed UX analysis platform** turning screenshots, OCR, interaction telemetry, and visual heuristics into structured findings (Angular · Spring Boot gateway · FastAPI services).
- Moved heavy AI work off the request path into an **async job lifecycle** with queueing, polling, and recovery; secured the gateway with Keycloak OAuth2/OIDC and JWT validation.
- Isolated LLM-inference, OCR, and image-processing workloads into fallback-aware FastAPI microservices on PostgreSQL, Redis caching, and Docker Compose.

PROJECTS

QuickFlow — AI Productivity Assistant

2025

- Built a full-stack AI productivity assistant for meeting summaries, structured minutes, and professional email generation.
- Developed Spring Boot backend services and Angular frontend workflows; integrated secure identity management with Keycloak and social login support.
- Designed reusable communication automation flows, keeping authentication and workflow reliability as first-class concerns.

AI Image Vectorization Pipeline

2025

- Designed a Python AI image-generation and vectorization pipeline for apparel design workflows under **8GB VRAM constraints**.
- Automated preprocessing, SAM segmentation, Oklab color simplification, and SVG tracing into repeatable vector-ready outputs.
- Treated hardware limits and repeatability as design constraints, with VRAM flushing and batch checkpointing for recovery.

Machine Learning / AI Data Projects

2024 – 2025

- Play Store Predictive Analysis:** trained deep-learning models on a **3M-row dataset** to predict app installs and extract application performance insights.
- Soil Nutrient Prediction:** built a reproducible Python pipeline for satellite/environmental data cleaning, feature engineering, and multi-output LightGBM modeling.

SKILLS

Programming: Python, Java, C/C++, C#, TypeScript/JavaScript, SQL

Agentic AI & LLMs: LangGraph, LangChain, LiteLLM, local LLMs (Ollama), RAG pipelines, reranking (TEI), semantic caching, faithfulness checks, tool-calling agents, MCP servers, prompt engineering

Machine Learning & Data: NumPy, pandas, Pydantic, scikit-learn, PyTorch, Keras, LightGBM, data preprocessing, feature engineering

Backend & Security: Spring Boot, FastAPI, Flask, Keycloak, OAuth2/OIDC, JWT, PostgreSQL, Redis, Qdrant, Celery, SSE

Automation, Vision & Media: Playwright browser automation, selector fallbacks, retry logic, queue orchestration, Segment Anything (SAM), VTracer/SVG tracing, Oklab quantization, OpenCV/Pillow, FFmpeg/Whisper

Frontend & Delivery: Angular, React, HTMX, Streamlit, Gradio, Docker & Compose, Linux CLI, Git/GitHub/GitLab, Postman, caching, fallback-aware workflows

CERTIFICATIONS

NVIDIA DLI: [Prompt Engineering for LLMs](#) · [RAG Agents with LLMs](#) · [AI Cybersecurity Pipelines](#) · [Transformer NLP](#) · [Diffusion Models](#) · [Real-Time Video AI](#) · [Predictive Maintenance](#) · [Fundamentals of Deep Learning](#)

DeepLearning.AI: [Neural Networks and Deep Learning](#) · [Supervised Machine Learning](#) | **Google Cloud:** [Introduction to Generative AI](#)

Competition: [IndabaX Tunisia 2025 Challenge 1](#) — top 50% (31 / 69), 20-hour applied-ML hackathon on 4G/5G telecom KPI anomaly detection

EDUCATION

École Nationale d'Ingénieurs de Sousse (ENISO)

Sep 2024 – Present

Engineering Degree in Software Engineering

Preparatory Institute for Engineering Studies of Monastir

Sep 2022 – Jun 2024

Mathematics & Physics Preparatory Cycle · National Rank: 509 / 1800

LANGUAGES

Arabic: Native | English: C1 | French: B2 | German: A1